

Extension Complexity

Def'n: The extension complexity of a polytope P is the minimum number of inequalities needed to describe a polytope Q s.t. P is an affine map of Q .

$$P = \{x : Ax \leq b\}, \quad Q = \{(x, y) : Cx + Dy \leq d\}$$

$$\text{s.t. } x \in P \iff \exists y \text{ s.t. } (x, y) \in Q$$

1. By ellipsoid alg; equiv of SEP & OPT, a LP is eff solvable iff $\exists$ polynomial separation oracle.
2. If P has poly-size extension, then OPT problem is eff. solvable, and if OPT is not solvable eff, then P cannot have poly-size ext compl.
3. However, above statement is not "iff"

Thm: [Thomas Rothvorn, '17] The ext. compl of the perfect matching polytope for the complete graph K_n , is $2^{\Omega(n)}$.

Nonnegative Rank and Slack

$$P = \text{conv} \{x_1, \dots, x_v\} = \{x : Ax \leq b\}, \quad m = \# \text{ rows of } A$$

S - slack matrix

$$S_{ij} \triangleq b_i - a_i^T x_j, \quad \text{where } a_i^T \text{ is } i^{\text{th}} \text{ row of } A \\ x_j \text{ is } j^{\text{th}} \text{ vertex of } P$$

Thm: (Yannakakis, '91)

The min comp (# inequalities) of an extended formulation for P is exactly equal to the non-negative rank of $S = \text{rank}_+(S), \text{rk}_+(S)$.

Def'n: Given a matrix M , $m \times r$, non-neg entries

$$\text{rk}_+(M) = \min \{r : M = UV^T, \quad U \in \mathbb{R}_{\geq 0}^{m \times r}, \quad V \in \mathbb{R}_{\geq 0}^{r \times r}\}$$

$$M = \begin{bmatrix} \vdots \end{bmatrix} \begin{bmatrix} \vdots \end{bmatrix}^T$$

$$M = \begin{pmatrix} 11 & 00 \\ 01 & 10 \\ 00 & 11 \\ 10 & 01 \end{pmatrix}$$

$$\text{rk}(M) = 3, \quad \text{rk}_+(M) = 4.$$

Proof — 1/3

$$S : S_{ij} = b_i - a_i^T x_j$$

$$r = \text{rk}_+(S) : S = UV^T, U \in \mathbb{R}_{\geq 0}^{m \times r}, V \in \mathbb{R}_{\geq 0}^{r \times n}$$

$$Q = \{(x, y) \in \mathbb{R}^{n+r} : \underbrace{Ax + Uy = b}_{m \text{ eq. in } n+r}, \underbrace{y \geq 0}_{r = \# \text{ ineq.}}\}$$

Claim: Q is an extended formulation of P

a. First, show that $\forall x_j \quad j=1, \dots, n$ ext pt of P

$$P \subseteq Q \left\{ \begin{array}{l} \exists y_j \text{ s.t. } y_j \geq 0 \text{ and } \underbrace{Ax_j + Uy_j = b.} \\ \text{Let: } y_j \in \mathbb{R}^r \text{ be the } j^{\text{th}} \text{ row of } V \text{ (i.e., } j^{\text{th}} \text{ col of } V^T) \\ \underbrace{A_i x_j + U_i y_j = b_i - S_{ij} + S_{ij} = b_i, \text{ Also: } y_j \geq 0,} \end{array} \right. \quad S_{ij} = \underbrace{\left[\begin{array}{c} u \\ y \end{array} \right]}_{U_i} \left[\begin{array}{c} \vdots \\ v^T \end{array} \right]$$

b. $\forall \hat{x} \notin P \Rightarrow \exists$ violated constraint, so say: $A_i \hat{x} > b_i$

but $U \geq 0$ and requirement $y \geq 0 \Rightarrow \nexists y \geq 0$ s.t. $\underline{A_i \hat{x}} + \underline{U_i y} = \underline{b_i}$

Proof — 2/3

We have shown: $rk_+(S) \geq \text{Ext Comp of } P$.

Now, ~~step~~ to show reverse inequality: suppose

$Q = \{(x, y) : Bx + Cy \leq d\}$ is an extended formulation of P with r inequalities.

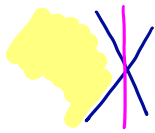
For each vertex x_j , $j=1, \dots, v$, fix y_j s.t.

$$(x_j, y_j) \in Q.$$

$$\text{Let } \boxed{v_j = d - Bx_j - Cy_j} \in \mathbb{R}_{\geq 0}^r$$

" Q is an ext form of P " : The system $Bx + Cy \leq d \Rightarrow Ax \leq b$

i.e., $\forall x, y$ if $Bx + Cy \leq d \Rightarrow Ax \leq b$.



We know: $\forall p \geq 0$ if x, y satisfy $Bx + Cy \leq d$

then they satisfy: $p^T (Bx + Cy) \leq p^T d$

Informal: LP Strong Duality says: $\exists p_i \geq 0$ s.t. $p_i^T B = A_i$, $p_i^T C = 0$, $p_i^T d = b_i$

Proof — 3/3

Use LP strong duality to show that:

$$\forall i, \exists p_i \geq 0 \text{ s.t. } p_i^T [B \ C] = [a_i \ 0 \ b_i]$$

$$b_i = \max_{x, y} : \begin{matrix} a_i^T x + 0^T y \\ \text{st: } Bx + Cy \leq d \end{matrix} \iff \min_{p_i} : \begin{matrix} p_i^T d \\ \text{st: } p_i^T [B \ C] = [a_i^T \ 0] \\ p_i \geq 0 \end{matrix}$$

By St. Duality, p_i^* is feasible:

$$p_i^* [B \ C] = [a_i^T \ 0]$$

$$p_i^T d = b_i.$$

Build matrix U by $U_i = p_i$

$$U_i V_j = U_i (d - Bx_j - Cy_j) = b_i - a_i^T x_j = S_{ij}.$$

Hence we have a non-veg fact of $S \Rightarrow r \geq \text{rk}_+(S)$